Four-color allocation comparison: cost-per-life efficiency + need plan

Coverage, service and lives outputs

Allocation_model_CostPerLife_Need.xlsx (cut = lambda x capacity x efficiency discount x need discount; efficiency = lives saved per US$1M in the dashboard, k = 4; need k = 4) | dashboard engine, 65 matched countries | November 11, 2026

Lives saved before any new money (Red → Blue): 392,003 over 2026-2030 (95% range 309,253 to 562,770). Extra deaths fall from 898,897 to 506,894. That is 132,432 more lives (95% range 79,289 to 180,556) than the original capacity-only plan saves from the same aid. With new money added (Green), extra deaths fall to zero under every plan.

Color	Scenario	Aid + new money, US$bn/year (82 countries)	Extra deaths 2026-2030 (95% range)
Red	Current aid only	25.914	898,897 (700k – 1,165k)
Yellow	Current aid + new money	90.618	199,893 (157k – 262k)
Green	Cost-per-life + need plan aid + new money	90.618	0 (0k – 0k)
Blue	Cost-per-life + need plan aid only	25.914	506,894 (352k – 665k)

Red and Blue have the same donor aid pool; Yellow and Green add the same domestic money. Red and Yellow are identical to the original capacity-only results; only Green and Blue use the cost-per-life + need plan.

Modeled service coverage

Service	Red	Yellow	Green	Blue
HIV treatment (ART)	76.77%	79.46%	80.71%	79.32%
PMTCT	85.60%	88.54%	89.69%	88.41%
Drug-susceptible TB treatment	77.89%	79.37%	79.69%	77.74%
Drug-resistant TB treatment	35.30%	36.19%	36.30%	34.32%
Bednet use	23.80%	24.39%	24.57%	24.29%
Indoor spraying	1.11%	1.14%	1.16%	1.15%
Malaria testing / treatment	49.74%	50.45%	50.78%	50.64%
Routine immunization	81.33%	83.14%	83.44%	82.51%

Extra deaths 2026-2030 by disease

Disease	Red	Yellow	Green	Blue
HIV	250,172	78,227	0	87,573
TB	343,944	60,969	0	366,402
Malaria	96,226	26,682	0	8,163
Immunization	208,555	34,015	0	44,756
Total	898,897	199,893	0	506,894

Coverage is weighted separately within each service across 65 matched countries. Deaths are relative to historical (2021-2023) funding and include HIV deaths from new infections; lower is better. The dashboard models losses and replacement up to historical coverage; it does not estimate expansion from money above that baseline.

All six pairwise comparisons

Second scenario minus first. Coverage in percentage points; lives saved = deaths in the first scenario minus deaths in the second. Positive means the second color is better.

Service (pp)	Red → Yellow	Red → Blue	Yellow → Green	Blue → Green	Blue → Yellow	Red → Green
HIV treatment (ART)	+2.6920	+2.5558	+1.2475	+1.3837	+0.1362	+3.9395
PMTCT	+2.9372	+2.8011	+1.1462	+1.2823	+0.1361	+4.0834
Drug-susceptible TB treatment	+1.4769	–0.1504	+0.3145	+1.9418	+1.6273	+1.7914
Drug-resistant TB treatment	+0.8860	–0.9789	+0.1109	+1.9758	+1.8649	+0.9970
Bednet use	+0.5885	+0.4916	+0.1880	+0.2850	+0.0969	+0.7765
Indoor spraying	+0.0296	+0.0387	+0.0182	+0.0092	–0.0090	+0.0478
Malaria testing / treatment	+0.7134	+0.8997	+0.3313	+0.1449	–0.1863	+1.0447
Routine immunization	+1.8106	+1.1763	+0.2990	+0.9333	+0.6343	+2.1096

Lives saved 2026-2030	Red → Yellow	Red → Blue	Yellow → Green	Blue → Green	Blue → Yellow	Red → Green
HIV	+171,946	+162,599	+78,227	+87,573	+9,346	+250,172
TB	+282,975	–22,458	+60,969	+366,402	+305,432	+343,944
Malaria	+69,544	+88,063	+26,682	+8,163	–18,519	+96,226
Immunization	+174,540	+163,799	+34,015	+44,756	+10,741	+208,555
Total	+699,004	+392,003	+199,893	+506,894	+307,001	+898,897
95% range	+544k to +907k	+309k to +563k	+157k to +262k	+352k to +665k	+178k to +446k	+700k to +1,165k

Service trade-off. Reallocation alone (Red → Blue) saves 392k lives but lowers coverage of Drug-susceptible TB treatment, Drug-resistant TB treatment. TB deaths still rise (from 343,944 to 366,402) because the reallocation takes aid from large TB-burden middle-income countries, but by less than under the capacity-only or mortality plans, because the cost-per-life discount protects countries where TB aid saves lives cheaply. New domestic money restores these losses in Green.

How to interpret the four-color design

Red → Yellow isolates new domestic money under current aid allocations. Red → Blue isolates reallocation with no new money. Yellow → Green isolates reallocation with domestic money. Blue → Green isolates domestic money under reallocated aid. Red → Green is the combined change; it should not be attributed entirely to reallocation.

Effects depend on the order of the changes

New money and allocation interact. The two-order average below splits the Red → Green change between them. It is an accounting decomposition of these model scenarios, not evidence of causal policy effects.

Measure	New-money contribution	Reallocation contribution	Combined change, Red → Green
HIV treatment (ART) (pp)	+2.0379	+1.9017	+3.9395
PMTCT (pp)	+2.1097	+1.9736	+4.0834
Drug-susceptible TB treatment (pp)	+1.7093	+0.0821	+1.7914
Drug-resistant TB treatment (pp)	+1.4309	–0.4340	+0.9970
Bednet use (pp)	+0.4367	+0.3398	+0.7765
Indoor spraying (pp)	+0.0194	+0.0284	+0.0478
Malaria testing / treatment (pp)	+0.4292	+0.6155	+1.0447
Routine immunization (pp)	+1.3719	+0.7377	+2.1096
Lives saved 2026-2030	+602,949	+295,948	+898,897

New money = [(Yellow – Red) + (Green – Blue)] / 2. Reallocation = [(Blue – Red) + (Green – Yellow)] / 2.

The full domestic-money addition is not all converted by this dashboard

Scenario	New money, US$bn/year	Fills historical aid gap, US$bn/year	Above historical aid baseline, US$bn/year
Yellow	64.705	3.745	60.959
Green	64.705	4.716	59.989

Workbook-wide all-health dollars across 82 countries. The money above the historical aid baseline is not declared unspent: the loss-only engine does not estimate its additional coverage effects.

Annual service outputs

Service units lost relative to historical service coverage. These are losses, so lower is better. Units differ and overlap across services.

Service	Unit	Red losses	Yellow losses	Green losses	Blue losses
HIV treatment (ART)	ART patient-years	1,198,516	379,343	0	421,844
PMTCT	women receiving PMTCT	41,459	11,606	0	13,117
Drug-susceptible TB treatment	TB courses	154,204	27,073	0	167,151
Drug-resistant TB treatment	DR-TB courses	2,811	313	0	5,571
Bednet use	people using bednets	25,540,806	6,184,988	0	9,373,173
Indoor spraying	people protected by spraying	1,573,505	599,670	0	302,185
Malaria testing / treatment	malaria cases treated	2,797,233	885,420	0	400,327
Routine immunization	children immunized	1,841,868	261,042	0	814,831

Annual service units retained: all six comparisons

Service	Red → Yellow	Red → Blue	Yellow → Green	Blue → Green	Blue → Yellow	Red → Green
HIV treatment (ART)	+819,173	+776,672	+379,343	+421,844	+42,501	+1,198,516
PMTCT	+29,853	+28,342	+11,606	+13,117	+1,511	+41,459
Drug-susceptible TB treatment	+127,130	–12,947	+27,073	+167,151	+140,077	+154,204
Drug-resistant TB treatment	+2,498	–2,760	+313	+5,571	+5,258	+2,811
Bednet use	+19,355,818	+16,167,633	+6,184,988	+9,373,173	+3,188,185	+25,540,806
Indoor spraying	+973,835	+1,271,320	+599,670	+302,185	–297,485	+1,573,505
Malaria testing / treatment	+1,911,813	+2,396,906	+885,420	+400,327	–485,092	+2,797,233
Routine immunization	+1,580,826	+1,027,037	+261,042	+814,831	+553,790	+1,841,868

Mixed-service summary

Scenario	Service equivalents lost/year (millions; not unique people)	Matched countries with remaining gaps	Extra deaths 2026-2030
Red	33.150	65	898,897
Yellow	8.349	17	199,893
Green	0.000	0	0
Blue	11.498	65	506,894

Lives saved: this plan vs the other allocation rules

Same aid pool, same new money, same engine. Only the rule that divides the cut changes. Blue = reallocation alone; in Green every rule closes every matched country's gap, so the rules differ only in Blue.

Allocation rule	Zero-aid countries (82)	Blue: extra deaths	Blue: lives saved vs current aid (95% range)	Blue: lives saved vs capacity only (95% range)
Capacity only (Allocation)	12	639,326	+259,571 (+164k to +450k)	—
Capacity + disease need (Allocation_Need)	13	605,121	+293,776 (+195k to +484k)	+34,205 (+27k to +43k)
Capacity + mortality efficiency + need	15	571,719	+327,178 (+227k to +520k)	+67,607 (+53k to +86k)
Capacity + cost-per-life efficiency	16	541,578	+357,319 (+277k to +523k)	+97,748 (+48k to +141k)
Capacity + cost-per-life efficiency + need (this plan)	19	506,894	+392,003 (+309k to +563k)	+132,432 (+79k to +181k)

Blue coverage by rule	Capacity only	+ need	+ mortality + need	+ cost per life	+ cost per life + need
HIV treatment (ART)	79.28%	79.44%	79.55%	79.11%	79.32%
PMTCT	88.35%	88.53%	88.66%	88.18%	88.41%
Drug-susceptible TB treatment	77.12%	77.21%	77.28%	77.66%	77.74%
Drug-resistant TB treatment	33.82%	33.92%	33.96%	34.22%	34.32%
Bednet use	24.24%	24.26%	24.29%	24.27%	24.29%
Indoor spraying	1.15%	1.15%	1.15%	1.15%	1.15%
Malaria testing / treatment	50.60%	50.62%	50.66%	50.61%	50.64%
Routine immunization	82.43%	82.44%	82.53%	82.51%	82.51%

What drives the ranking (dashboard, median over countries)	HIV	TB	Malaria	Immunization
Lives saved per US$1M of aid lost	186	1,165	90	431
Cost per life saved (US$)	$5,375	$858	$11,051	$2,322

Median over countries where the disease is at least 5% of aid. A country's overall figure is these, for that country, weighted by its own aid mix (Data_CostPerLife).

Reading this page. Ranking countries by how cheaply the dashboard saves a life protects them better than ranking by mortality: with the need discount, it saves 132k more lives than the capacity-only plan and 65k more than the mortality version (95% range 23k to 99k). Most of the gain comes from TB, the dashboard's cheapest way to save a life. **Caution:** the coefficient is built from the same dashboard that scores the results, so the plan is designed to look good by this measure. Treat its lives saved as an upper bound; the mortality version is the more independent test.

Country distribution: cost-per-life + need plan vs capacity-only plan (Blue)

Where the efficiency and need discounts move aid, and the lives that follow (deaths 2026-2030, central). Five largest gains and five largest losses.

Country	Cost per life saved US$	TB share of aid	Aid 2023 US$m	Aid, capacity plan US$m	Aid, this plan US$m	Deaths, capacity plan	Deaths, this plan	Lives saved
Indonesia	$811	39%	1,513	692	1,029	148,584	86,964	+61,620
Philippines	$753	37%	268	0	120	123,275	67,581	+55,695
India	$1,781	45%	1,433	413	518	125,492	112,520	+12,973
Pakistan	$1,005	14%	944	896	909	19,313	14,067	+5,246
Nigeria	$1,263	10%	1,923	1,896	1,910	9,103	4,535	+4,569
Cambodia	$1,881	26%	204	182	176	4,930	6,229	–1,299
Nepal	$3,129	16%	279	250	229	2,088	3,537	–1,449
Kyrgyzstan	$8,865	40%	64	37	0	1,183	2,797	–1,614
Guatemala	$6,096	8%	176	84	0	2,446	4,747	–2,301
Vietnam	$1,563	22%	251	22	0	64,834	71,095	–6,261

Across all 82 workbook countries the cost-per-life + need plan moves US$650 m of aid relative to the capacity-only plan. 20 countries gain aid and 30 lose aid. These rankings depend on service mix and are not a ranking of unique people helped.

How the cost-per-life efficiency coefficient works

Change	Detail
The rule	Each country's cut = lambda x effective capacity, never more than its 2023 aid. Effective capacity = domestic capacity x efficiency discount x need discount. Lambda is solved so the cuts add up exactly to the OECD-projected total cut (US$4,716 m); the 2026 aid pool stays US$25,914 m. Countries that can raise more lose more; countries where aid saves a life cheaply, and high-burden countries, are asked for less.
How the dashboard turns dollars into lives	For each country, every HIV, TB, malaria and immunization aid line is cut by 10% (no government replacement) and the engine reports aid lost and extra deaths 2026-2030 by disease. Its calculation uses (1) the country's own mix of aid across services (IHME, by donor and channel); (2) unit costs: country-specific for ART and TB treatment, standard costs for bednets, spraying, malaria treatment, PMTCT and vaccines; (3) deaths averted per unit of service; (4) a coverage cap: donor money only buys people it actually covers (people covered x donor share), so where donor money exceeds that pool it saves fewer lives per dollar. Part of aid (systems, management, orphan support) buys no modelled service.
Efficiency coefficient (Data_CostPerLife, Efficiency)	Lives per US$1M = sum over the four diseases of (share of the country's aid) x (lives per US$1M for that disease). Cost per life saved = US$1M / lives per US$1M. Efficiency = lives per US$1M / the highest value (cheapest country = 1); countries not in the dashboard take their income-group average. Discount = 1 / (1 + 4 x efficiency). The per-disease shares and lives per US$1M are in the workbook, so each country's figure can be traced.
Workbook layout and checks	Overview, Headline, Allocation and Cut_Solver run this plan. Levers: Assumptions B26 (need k), B37 (efficiency k). Setting B37 to 0 reproduces the original need-adjusted plan exactly; both to 0 gives the capacity-only plan. Aid pool US$25,914 m; no formula errors. The per-disease breakdown adds back to the dashboard total exactly.
Dashboard run	Country cut rate = G/E (Red, Yellow) or L/E (Green, Blue); government replacement = 0 (Red, Blue), min(I, G)/G (Yellow), min(I, L)/L (Green), applied to every donor/channel/service cell. health_model.run_country with Scenario(announcement=True), Fiscal(custom, theta, cap_to_space=False, pro_rata, p75), 400 draws, seed 7. Ranges are 2.5th-97.5th percentiles of draw-by-draw differences.
Limitations	Circular by design: the coefficient comes from the same dashboard that scores the plan, so its lives saved are an upper bound. It inherits the dashboard's assumptions, e.g. TB treatment is its cheapest way to save a life, and it measures what is lost when aid is cut, not what extra money buys. The engine gives no credit for money above 2023 aid. Coverage and lives cover 65 countries; the workbook covers 82 and all health aid. Fixed-scenario illustrations, not forecasts.